

Equations, functions and graphs

Solving equations, what a function is, and the shapes you have to recognise on sight.

- 1 Linear equations and systems
 - 2 Quadratic equations in business
 - 3 What a function is
 - 4 Domain and range
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Linear equations and systems

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► **Watch:** First order equations · Why we do the same thing to both sides (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:solve-equations-inequalities/x2f8bb11595b61c86:linear-equations-variables-both-sides/v/why-we-do-the-same-thing-to-both-sides-multi-step-equations>)

► **Watch:** Systems, by substitution · Solving systems of equations with substitution (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:systems-of-equations/x2f8bb11595b61c86:solving-systems-of-equations-with-substitution/v/solving-systems-with-substitution>)

≡ **Read:** The method, written out · Multi-step equations review (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:solve-equations-inequalities/x2f8bb11595b61c86:linear-equations-parentheses/a/multi-step-equations-review>)

✍ **Practise:** First order equations · Equations with variables on both sides (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:solve-equations-inequalities/x2f8bb11595b61c86:linear-equations-variables-both-sides/e/linear_equations_3)

✍ **Practise:** Systems, by elimination · Systems of equations with elimination (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:systems-of-equations/x2f8bb11595b61c86:solving-systems-elimination/e/systems_of_equations_with_elimination_0.5)

FIRST ORDER

One unknown, first power. $ax + b = 0$ with $a \neq 0$.

A SYSTEM

Two equations that have to hold at the same time, for the same pair of values.

- 1 **Clear brackets, then fractions.** Multiply every term by the common denominator, the right-hand side included.
- 2 **Collect the unknown on one side, the numbers on the other.**
- 3 **Divide by the coefficient.**

$$ax + b = 0 \quad \Rightarrow \quad x = \frac{-b}{a}$$

- 4 **For a system, eliminate.** Scale one equation so a letter cancels when you add the two.
- 5 **Or substitute.** Make one equation say $x = \text{something}$, and put that into the *other* one.
- 6 **Check the pair in both equations.** One equation cannot catch an error.

THREE OUTCOMES

One solution. None, when you reach something false. Infinitely many, when you reach something always true. None of the three is a mistake.

First order equations

$$x = \frac{-b}{a}$$

WORKED: $3x + 1 = 5x + 7$

$$3x + 1 = 5x + 7$$

Collect the x terms on one side, the numbers on the other.

$$-2x = 6$$

$$3x - 5x = -2x \text{ and } 7 - 1 = 6.$$

$$x = -3$$

Divide by -2 .

Systems of first order equations

THE SAME SYSTEM, BOTH WAYS

$$2x + y = 18 \text{ and } -x + 5y = 2$$

The system to solve.

$$-2x + 10y = 4$$

Method 1: multiply the second equation by 2, then add it to the first.

$$11y = 22, \text{ so } y = 2 \text{ and } x = 8$$

The x terms cancel.

$$x = 5y - 2$$

Method 2: isolate x in the second equation instead.

$$2(5y - 2) + y = 18$$

Substitute into the first.

$$11y = 22, \text{ so } y = 2 \text{ and } x = 8$$

The same answer, as it must be.

WATCH OUT

- Multiplying an equation in a system means multiplying its right-hand side too.
- $-(x - 4) = -x + 4$. Losing that second sign is the commonest slip here.

- Check your pair in both equations. One is not enough.

Quadratic equations in business

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▶ **Watch:** The formula · The quadratic formula (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:quadratic-formula-a1/v/using-the-quadratic-formula>)

▶ **Watch:** What the discriminant tells you · Using the quadratic formula: number of solutions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:quadratic-formula-a1/v/quadratic-formula-3>)

≡ **Read:** The discriminant · Discriminant review (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:quadratic-formula-a1/a/discriminant-review>)

✍ **Practise:** Using the formula · Quadratic formula (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:quadratic-formula-a1/e/quadratic_equation)

✍ **Practise:** Reading the discriminant · Number of solutions of quadratic equations (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:quadratic-formula-a1/e/determine-the-number-of-solutions-of-a-quadratic-equation>)

1 Everything to one side.

$$ax^2 + bx + c = 0, \quad a \neq 0$$

2 Read off a, b and c with their signs. The signs are where the marks go.

3 Work out the discriminant first.

$$\Delta = b^2 - 4ac \quad \Delta > 0: \text{two roots} \quad \Delta = 0: \text{one} \quad \Delta < 0: \text{none in } \mathbb{R}$$

4 Factor if it factors, otherwise use the formula.

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

5 In a business question, break-even is where revenue meets cost. $R(q) = C(q)$ rearranges to a quadratic you already know how to solve.

6 Throw away a root the situation forbids. A negative quantity or a negative price is not an answer.

$$x_1 = \frac{-b + \sqrt{\Delta}}{2a} \quad x_2 = \frac{-b - \sqrt{\Delta}}{2a}$$

WORKED: $3x^2 + 3x - 18 = 0$

$$\Delta = 3^2 - 4(3)(-18) = 225$$

Positive, so two solutions.

$$x = \frac{-3 \pm 15}{6}$$

$$\sqrt{225} = 15.$$

$$x = 2 \text{ or } x = -3$$

The two roots.

$$3x^2 + 3x - 18 = 3(x + 3)(x - 2)$$

It also factors directly, which is faster when the numbers are kind.

The application: Berger Tools

THE SOLUTION

$$C(q) = 6q + 80\,000$$

Total cost: variable plus fixed.

$$R(q) = 10q$$

Total revenue: price times quantity.

$$P(q) = R(q) - C(q) = 4q - 80\,000$$

Profit is the difference of the two functions.

$$60\,000 = 4q - 80\,000$$

Set the profit to the target.

$$q = 35\,000 \text{ units}$$

Which is the answer to the business question, not just to the algebra.

WATCH OUT

- $\Delta < 0$ means no solution in $\mathbb{R}$. Write that; it is a complete answer.
- In $-4ac$ with c negative the two signs make a plus. Most wrong discriminants come from here.
- Move everything to one side before reading off a , b and c .

What a function is

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- ▶ **Watch:** What counts as a function · Recognizing functions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:recognizing-functions/v/graphical-relations-and-functions>)
- ▶ **Watch:** Evaluating a function · Worked example: Evaluating functions from equation (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:evaluating-functions/v/understanding-function-notation-example-1>)
- ▶ **Watch:** Lines, slope and intercept · Worked examples: slope-intercept intro (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:forms-of-linear-equations/x2f8bb11595b61c86:intro-to-slope-intercept-form/v/slope-intercept-intro-examples>)
- ✍ **Practise:** Evaluating · Evaluate functions (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:evaluating-functions/e/functions_1)
- ✍ **Practise:** The vertical line test · Recognize functions from graphs (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:recognizing-functions/e/recog-func-2>)
- ✍ **Practise:** Reading a line off a graph · Slope-intercept equation from graph (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:forms-of-linear-equations/x2f8bb11595b61c86:writing-slope-intercept-equations/e/slope-intercept-equation-from-graph>)

FUNCTION

A rule that gives each allowed input **exactly one** output. That is the whole definition.

$f(x)$

The output at x . It is not f times x .

FOUR REPRESENTATIONS

A formula, a table, a graph, a sentence. The same function, four ways, and you are expected to move between them.

VERTICAL LINE TEST

If a vertical line meets the graph twice, it is not a function: that input would have two outputs.

- 1 **To evaluate, replace every x by the whole input.** Brackets included: $f(a + 1)$ needs $(a + 1)$ everywhere.
- 2 **Read a table exactly, a graph approximately.** A graph gives you the shape and the trend, not the fourth decimal.

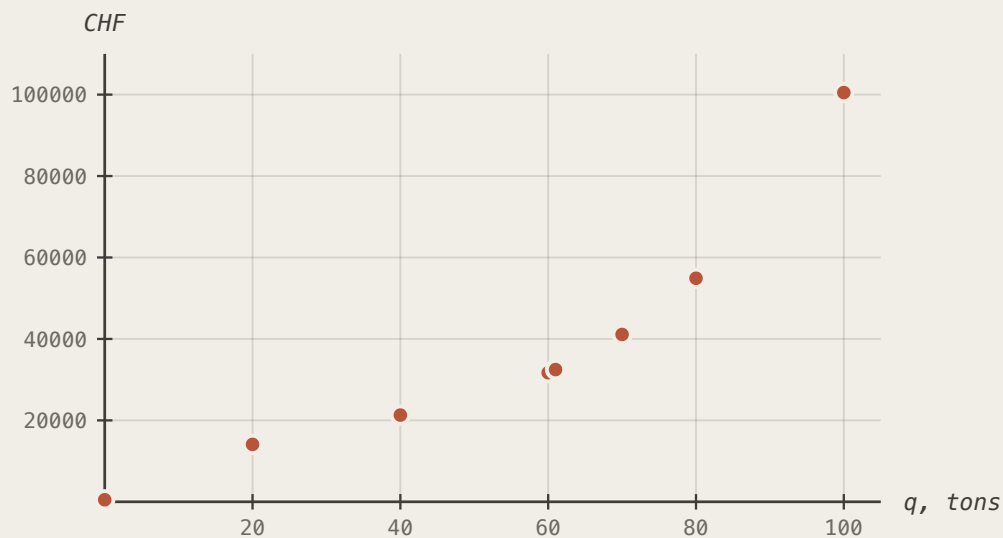
- 3 Slope is rise over run, and it is the same wherever you measure it.

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

- 4 To sketch a line, plot two points and join them. The intercepts are usually the two easiest.
- 5 A vertical line has no slope. $x = k$ is not a function, and its slope is undefined rather than zero.

The course definition

The worked case: Rivella Coffee



The Rivella Coffee cost table, drawn. Cost rises with quantity, and faster at the top end.

The four representations

EVALUATING, AND WHY BRACKETS MATTER

$$h(x) = 2$$

A constant function: every input gives 2.

$$h(10) = 2, h(-213) = 2, h(x + 3) = 2$$

The third one makes the point.

$$f(x) = x^2 - 3x \text{ at } x = a + 1$$

Replace every x by $(a + 1)$, brackets included.

$$(a + 1)^2 - 3(a + 1) = a^2 - a - 2$$

Expand afterwards.

WATCH OUT

- $f(x)$ is one number, the output. It is not f times x .
- $f(a + 1)$ is not $f(a) + 1$.
- A vertical line meeting the graph twice means it is not a function: that input would have two outputs.

Graphs, and the standard the course sets

Lines

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$



A line through two points. The slope is the rise over the run, and it is the same wherever you measure it.

The standard the course sets

Sketching by hand

WATCH OUT

- Slope is $\frac{\Delta y}{\Delta x}$. Subtract the coordinates in the same order top and bottom.
- $x = k$ is vertical and has undefined slope, which is not slope zero.
- An unlabelled sketch earns no marks even when the curve is right.

Domain and range

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► **Watch:** What is the domain of a function? (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:introduction-to-the-domain-and-range-of-a-function/v/domain-of-a-function-intro>)

► **Watch:** What is the range of a function? (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:introduction-to-the-domain-and-range-of-a-function/v/range-of-a-function>)

✍ **Practise:** Finding a domain · Determine the domain of functions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:determining-the-domain-of-a-function/e/domain-of-algebraic-functions>)

✍ **Practise:** Reading both off a graph · Domain and range from graph (https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:introduction-to-the-domain-and-range-of-a-function/e/domain_and_range_0.5)

DOMAIN

Every input the rule accepts.

RANGE

Every output that actually occurs. Usually smaller than you expect.

- 1 **Start by assuming every real number is allowed.** Then take away what is forbidden.
- 2 **A denominator may not be zero.** Set it to zero, solve, and exclude those values.
- 3 **An even root may not have a negative underneath.** $\sqrt{x - 3}$ needs $x \geq 3$.
- 4 **A logarithm needs a strictly positive argument.**
- 5 **Solve each condition, then keep everything that survives all of them.**
- 6 **Write it whichever way you like.** An inequality, a set or an interval. All three are the same answer.
- 7 **From a graph, read both straight off the axes.** The domain is how far the curve reaches left and right, the range how far it reaches up and down.

WORKED: $F(X) = X \div (X - 2)$

$$f(x) = \frac{x}{x - 2}$$

A quotient, so the denominator is the risk.

$$x - 2 = 0 \rightarrow x = 2$$

Find where the denominator vanishes.

every real number except 2

That is the domain.

A SECOND: $F(X) = X \div (X^2 - X - 2)$

$$x^2 - x - 2 = 0$$

Same method, harder denominator.

$$(x - 2)(x + 1) = 0$$

Factor it.

every real number except 2 and -1

Two values removed, not one.

When two things can go wrong at once

Three ways to write it, and the range

WATCH OUT

- A root needs ≥ 0 ; a logarithm needs > 0 , strictly.
- Simplifying first can hide a restriction. Take the domain from the formula as given.
- Context can shrink a domain the algebra allows: 7.3 employees is not an input.

Six families of function

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- ▶ **Watch:** Parabolic functions · Intro to parabolas (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:intro-parabolas/v/parabolas-intro>)
- ▶ **Watch:** Graphs of absolute value functions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:absolute-value-piecewise-functions/x2f8bb11595b61c86:graphs-of-absolute-value-functions/v/shifting-absolute-value-graphs>)
- ▶ **Watch:** Graphs of rational functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:rational-functions/x9e81a4f98389efdf:graphs-of-rational-functions/v/finding-asymptotes-example>)
- ✍ **Practise:** Parabolas intro (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:intro-parabolas/e/parabolas-intro>)
- ✍ **Practise:** Case-defined functions · Evaluate piecewise functions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:absolute-value-piecewise-functions/x2f8bb11595b61c86:piecewise-functions/e/evaluating-piecewise-functions>)

THE SIX FAMILIES

Constant, linear, quadratic, polynomial, rational, absolute value. Recognising the family from the formula tells you the shape before you draw anything.

- 1 **Look at the formula, not at the picture.** The form is what names the family.
- 2 **Linear: one x, first power.** A straight line.
- 3 **Quadratic: there is an x^2 .** A parabola, opening upwards when $a > 0$.
- 4 **Rational: an x in a denominator.** So find the domain before anything else.
- 5 **Absolute value: distance from zero, so never negative.**

$$|x| = x \text{ if } x \geq 0, \quad -x \text{ if } x < 0$$

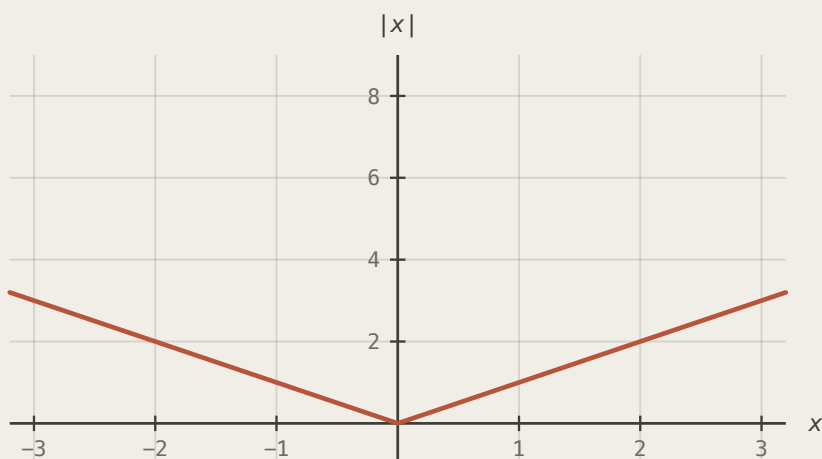
- 6 **Case-defined: several rules over several intervals.** Graph each piece only over its own interval, and check what happens at the joins.

The six families

FAMILY	DEFINITION	WHAT TO NOTICE
Constant	$h(x) = c$	every input gives the same output
Power	$f : \mathbb{R}^+_{\neq 0} \rightarrow \mathbb{R}^+, x \rightarrow x^n$	the variable is the base
Parabolic	$f(x) = ax^2 + bx + c$	three constants, opens up or down
Polynomial	$a_n x^n + \dots + a_1 x + a_0$	n is the degree
Rational	$f(x) = \frac{P(x)}{Q(x)}$	a quotient of polynomials
Absolute value	$ x $	two behaviours, one each side of 0

Absolute value

$$|x| = x \text{ if } x \geq 0, \quad -x \text{ if } x < 0$$



The absolute value function. Two straight pieces meeting at the origin, one for each sign of x .

Case-defined functions

GRAPHED

$$f(x) = x \text{ for } 0 \leq x < 3$$

First piece: the line $y = x$.

$$f(x) = x - 1 \text{ for } 3 \leq x < 5$$

Second piece: the same slope, shifted down.

$$f(x) = 4 \text{ for } 5 \leq x < 7$$

Third piece: constant.

WATCH OUT

- x^2 is a power function and 2^x is exponential. The variable is in a different place.
- A case-defined function is one function, not three. It has one domain and one range.
- At a boundary, only one case applies. The inequality signs tell you which.

Exponential functions

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- ▶ **Watch:** What makes it exponential · Exponential vs. linear growth (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:exponential-growth-decay/x2f8bb11595b61c86:exponential-vs-linear-growth/v/exponential-growth-functions>)
- ▶ **Watch:** Where e comes from · e and compound interest (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:e/v/e-through-compound-interest>)
- ▶ **Watch:** e as a limit · e as a limit (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:e/v/e-as-limit>)
- ✍ **Practise:** Growth and decay in context · Exponential expressions word problems (numerical) (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:exponential-growth-decay/x2f8bb11595b61c86:exponential-expressions/e/exponential-expressions-word-problems>)
- ✍ **Practise:** The two shapes · Graphs of exponential growth (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:exponential-growth-decay/x2f8bb11595b61c86:graphs-of-exponential-growth/e/graphing-exponential-growth-intro>)

EXPONENTIAL FUNCTION

$f(x) = b^x$ with $b > 0$ and $b \neq 1$. The variable is in the exponent, and that is what makes it exponential.

GROWTH OR DECAY

Growth when $b > 1$, decay when b is between 0 and 1.

- 1 Growing by $p\%$ each period multiplies by the same number each period.

$$\times \left(1 + \frac{p}{100}\right) \text{ each period}$$

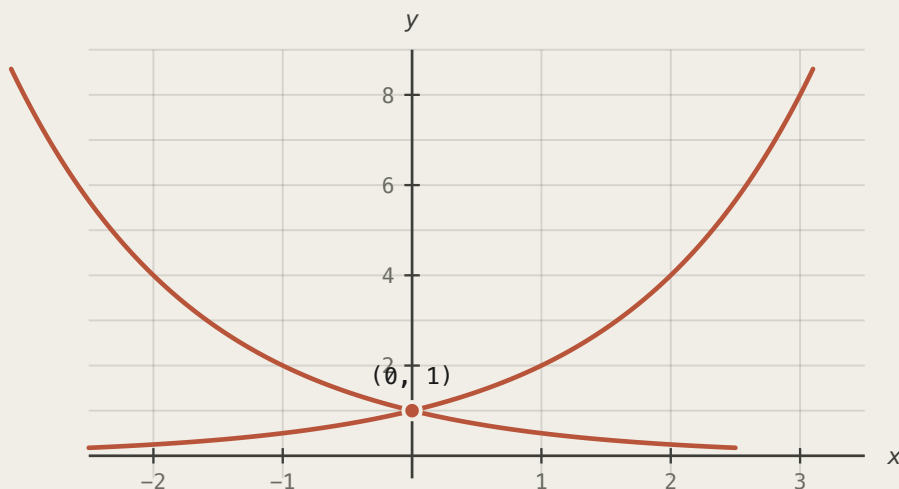
- 2 Compound interest is the case you will meet most.

$$S = P(1 + r)^n$$

P the principal, r the rate per period, n the number of periods.

- 3 Read the period off the question. 8% a month and 8% a year are not the same number.
- 4 e is a limit, roughly 2.718. $\exp(x)$ and e^x are the same thing.

- 5 A quantity that multiplies by a fixed factor each step is exponential. One that adds a fixed amount each step is linear. That distinction is most of the exam questions.



Growth and decay. 2^x doubles every step, 0.5^x halves, and both pass through (0, 1).

Compound interest, which is where it comes from

$$S = P(1 + r)^n$$

WORKED: 10 000 CHF AT 2% FOR THREE YEARS

$$S(3) = 10\,000 \times (1.02)^3$$

The multiplier is 1.02, not 0.02.

$$= 10\,612 \text{ CHF}$$

Rounded to the franc, to the franc.

The number e

SUBSCRIBERS GROWING 8% A MONTH FROM 500

$$N(t) = 500 \cdot 1.08^t$$

Same shape as compound interest.

$$N(6) = 500 \cdot 1.08^6$$

Six months.

$$= 793 \text{ subscribers}$$

$$1.08^6 \approx 1.5869.$$

WATCH OUT

- The multiplier for a rise of r per cent is $1 + \frac{r}{100}$, not $\frac{r}{100}$.
- $b \neq 1$ is part of the definition: 1^x is constant, not exponential.
- An exponential never reaches zero, so a decay question never answers "it becomes zero".

Logarithmic functions

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- ▶ **Watch:** What a logarithm is · Introduction to logarithms (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-intro/v/logarithms>)
- ▶ **Watch:** The shape of the graph · Graphs of logarithmic functions (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:transformations/x2ec2f6f830c9fb89:log-graphs/v/comparing-exponential-logarithmic-functions>)
- ≡ **Read:** The definition, written out · Intro to logarithms (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-intro/a/intro-to-logarithms>)
- ✍ **Practise:** Converting between the two forms · Relationship between exponentials & logarithms (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-intro/e/understanding-logs-as-inverse-exponentials>)
- ✍ **Practise:** Evaluating without a calculator · Evaluate logarithms (https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-intro/e/logarithms_1)
- ✍ **Practise:** Graphing · Graphs of logarithmic functions (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:transformations/x2ec2f6f830c9fb89:log-graphs/e/graphs-of-exponentials-and-logarithms>)

LOGARITHM

$$y = \log_b x \quad \text{if and only if} \quad b^y = x$$

The logarithm answers: to what power do I raise the base to get this number?

THE TWO ON YOUR CALCULATOR

log is base 10. ln is base e.

- 1 **To convert, ask what power the base is raised to.** Reading the statement the other way round is most of the work.
- 2 **Each undoes the other.**

$$\log_b(b^x) = x \quad b^{\log_b x} = x$$
- 3 **Without a calculator, write the argument as a power of the base.** $\log_2 32$: 32 is 2^5 , so the answer is 5.
- 4 **Only a strictly positive argument has a logarithm.** Check that before you do anything else.

- 5 To solve **b** to the power **x** equals **k**, take the **log of both sides**. The power rule then brings the **x** down where you can reach it.

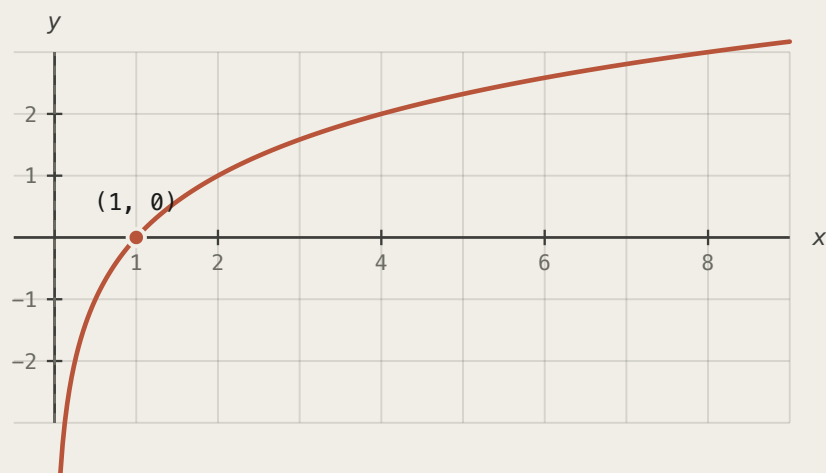
$$y = \log_b x \quad \text{if and only if} \quad b^y = x$$

$$\log_b(b^x) = x \quad b^{\log_b x} = x$$

Converting between the two forms

Conversions	
EXPONENTIAL FORM	LOGARITHMIC FORM
$5^2 = 25$	$\log_5 25 = 2$
$3^4 = 81$	$\log_3 81 = 4$
$10^0 = 1$	$\log_{10} 1 = 0$
$10^3 = 1000$	$\log_{10} 1000 = 3$
$64^{1/2} = 8$	$\log_{64} 8 = \frac{1}{2}$
$2^{-4} = \frac{1}{16}$	$\log_2 \frac{1}{16} = -4$

The two special logarithms



$y = \log_e x$. Defined only for $x > 0$, crossing at (1, 0), falling without limit towards the axis.

WORKED, WITHOUT A CALCULATOR

$$\log 100 = 2$$

Because $100 = 10^2$.

$$\ln 1 = 0$$

Because $e^0 = 1$.

$$\log 0.1 = -1$$

Because $0.1 = 10^{-1}$.

$$\ln e^{-1} = -1$$

The two undo each other.

WATCH OUT

- $\log x$ needs $x > 0$, strictly. No power of a positive base gives zero or a negative.
- The graph falls without limit as x approaches 0. It never crosses the vertical axis.
- $\log(a + b)$ does not break up. There is no rule for the logarithm of a sum.

Week 1 review quiz

Seven questions in the format of the Cyberlearn quiz: four answers and "none of the above". The wrong options are the answers you get from the usual slips, so read them before choosing.



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